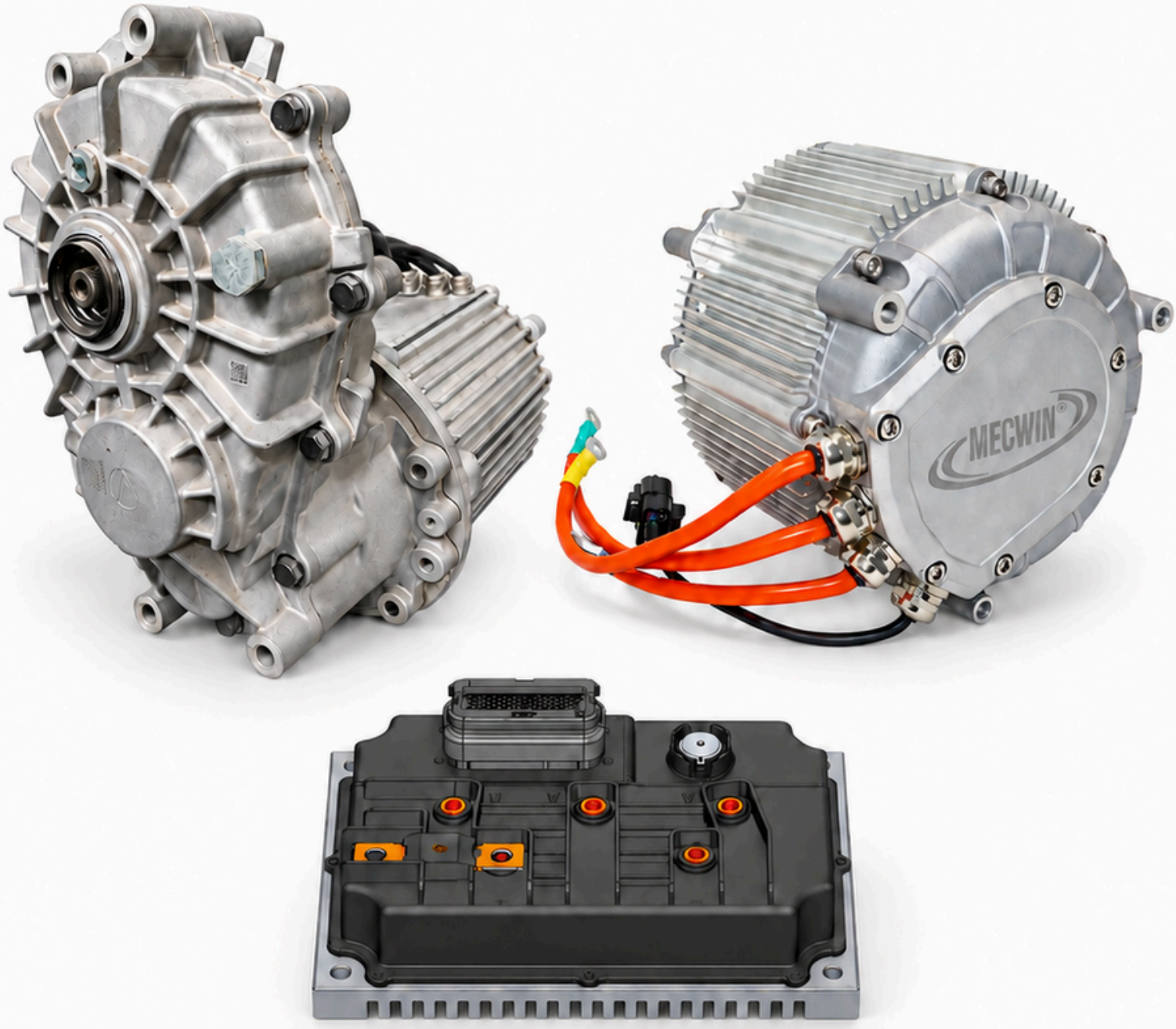
MECWIN®

EMPOWERING FUTURE

MID- DRIVE MOTORS & CONTROLLER



COMPANY OVERVIEW

About Us

Mecwin Group is a Bengaluru-based engineering and manufacturing company specializing in **motors, controllers, powertrains, wiring harnesses, SCADA automation and IoT solutions.** We develop customized solutions for **electric mobility, industrial automation, agriculture, renewable energy, Oil & Gas, aerospace, defence** and other demanding applications.

Our 50,000 sq. ft. manufacturing facility is equipped with modern machinery, engineering laboratories and advanced testing facilities. With strong in-house engineering capabilities, we manage the complete product journey from design and prototyping to testing, validation and production. We work closely with OEMs to provide high-performance, reliable and application-specific solutions.

Our growth is driven by innovation, engineering excellence, quality, integrity and long-term partnerships with our customers, suppliers and employees.

Vision

To empower a future where technology, people and the planet progress together.

Mission

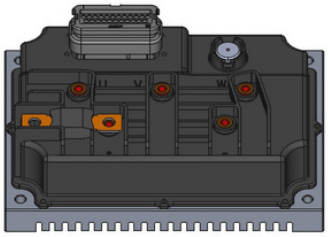
To become the preferred global technology partner for high-efficiency motors, controllers and powertrains—from agriculture to aviation—by delivering innovation, quality, reliability, on-time service and environmentally responsible manufacturing.

Core Values

Customer Success | Innovation | Excellence | Integrity | People & Partnerships | Environmental Responsibility.

EV drive product range

A clear view of the motor,controller and gear box options in this reference range.



Drive controller

72 V controller with CAN / YXT communication, forced-air cooling and IP67 ingress protection.



Hairpin-wound PMSM motors

Flat-wire hairpin-wound 5 kW and 7 kW model families with magnetic encoder feedback and NTC temperature signals.



Motor + gearbox

5 kW and 7 kW geared variants, with listed peak torque of 225 N.m and 285 N.m respectively.

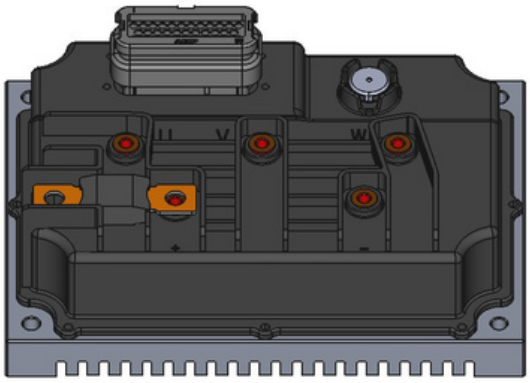
How to select a configuration

Confirm vehicle duty, battery voltage, continuous and peak load, gearing, mounting and cooling requirements. Match the motor and controller using the approved model-specific datasheets.

DRIVE CONTROL

Flat-Wire Drive Controller

Motor drive control for compatible electric vehicle powertrains.





Key features

CAN / YXT communication • Temperature-based derating Forced-air cooling • IP67 ingress protection

Technical specifications

Rated power	5-10 kW
Peak power	14-24 kW
Rated voltage	72 V
Voltage range	50-89 V
Maximum bus current	210 A
Peak phase current	500 A
Communication	CAN / YXT
Derating temperature	105 °C
Protection temperature	115 °C
Cooling	Forced air
Ingress protection	IP67
Mounting size	200 × 140 × 70 mm
Ambient temperature	-25 to 50 °C
Weight	1.4 kg

TRACTION MOTORS

5 kW Hairpin-Wound PMSM Motor

Permanent magnet synchronous motor for compatible electric mobility platforms.



PRODUCT OVERVIEW

This hairpin-wound flat-wire PMSM model offers a listed peak torque of 75 N·m and peak speed of 8,000 RPM. It uses magnetic encoder feedback and an NTC temperature signal.

KEY FEATURES

• Listed maximum efficiency >95% • Forced-air cooling
• IP67 ingress protection
• Temperature protection thresholds

Technical specifications

Rated power	5-9 kW
Peak power	13-20 kW
Peak speed	8,000 RPM
Peak torque	75 N.m
Peak phase current	500 A
Maximum efficiency	>95%
Temperature signal	NTC
Derating / protection temperature	140 °C / 160 °C
Cooling	Forced air
Encoder signal	Magnetic encoder
Ingress protection	IP67
Dimensions	Ø180 × 154
Weight	12 kg

Applications: Electric cargo three-wheelers, passenger three-wheelers and heavy-duty electric three-wheelers.

TRACTION MOTORS

7 kW Hairpin-Wound PMSM Motor

Permanent magnet synchronous motor for compatible electric mobility platforms.



PRODUCT OVERVIEW

This hairpin-wound flat-wire PMSM model offers a listed peak torque of 95 N·m and peak speed of 8,000 RPM. It uses magnetic encoder feedback and an NTC temperature signal.

KEY FEATURES

- Listed maximum efficiency >95%
- Forced-air cooling
- IP67 ingress protection
- Temperature protection thresholds

Technical specifications

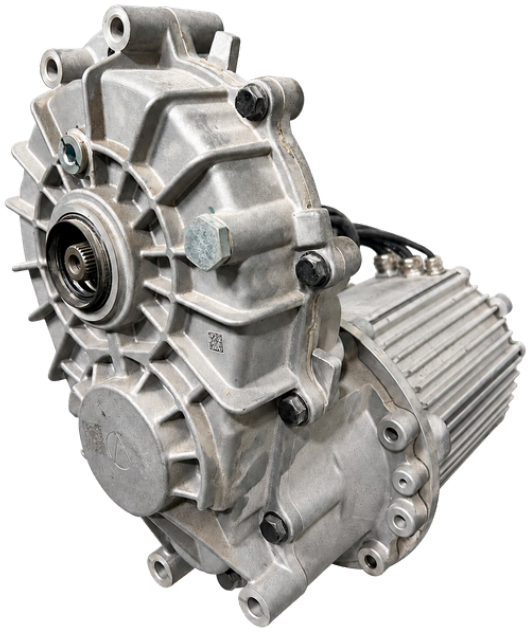
Rated power	7-10 kW
Peak power	14-24 kW
Peak speed	8,000 RPM
Peak torque	95 N·m
Peak phase current	500 A
Maximum efficiency	>95%
Temperature signal	NTC
Derating / protection temperature	140 °C / 160 °C
Cooling	Forced air
Encoder signal	Magnetic encoder
Ingress protection	IP67
Dimensions	Ø180 × 174
Weight	14 kg

Applications: Electric cargo three-wheelers, passenger three-wheelers and mini four-wheel EVs.

GEARED DRIVE

Hairpin-Wound PMSM + Gearbox

Flat-wire hairpin-wound motors with geared configurations for vehicle integration.



The reference lists a 1:3 gear ratio for both variants. Confirm the ratio convention and whether speed, torque and efficiency refer to the motor or gearbox output.

Variant comparison

Parameter	7 kW variant	5 kW variant
Rated power	7 kW	5 kW
Peak power	14 kW	10 kW
Peak speed	2,666 RPM	2,666 RPM
Peak torque	285 N·m	225 N·m
Maximum efficiency	>95%	>95%
Peak phase current	500 A	500 A
Gear ratio (as listed)	1:3	1:3
Temperature signal	NTC	NTC
Cooling	Forced air	Forced air
Encoder signal	Magnetic encoder	Magnetic encoder
Ingress protection	IP67	IP67
Derating temperature	140 °C	140 °C
Protection temperature	160 °C	160 °C
Dimensions	Ø180 × 230	Ø180 × 200
Weight	17.5 kg	15.5 kg

*Dimension units are not stated in the reference. Illustration is representative of the geared range.

APPLICATIONS & ENQUIRIES

Find the right drive solution

Select the configuration around your vehicle and operating requirements.

01 Electric cargo three-wheelers

For evaluation against payload, starting torque, route profile and operating duty.

02 Electric passenger three-wheelers

For evaluation against passenger capacity, speed targets and daily operating conditions.

03 Heavy-duty electric three-wheelers

Listed for the 5 kW motor family; confirm thermal limits and continuous-duty requirements.

04 Mini four-wheel EVs

Listed for the 7 kW motor family; confirm drivetrain compatibility and vehicle performance.

Share your requirements

Vehicle type • Battery voltage • Payload / gross vehicle weight • Target speed Gradeability • Duty cycle • Mounting space • Required interfaces

MECWIN Global Customers



International Compliance Support

- Products can bedesignedandvalidatedforapplicable:
- IEC and EN standards
- CE conformity requirements
- UL and CSA requirements
- RoHS and REACH compliance
- ATEX requirements for hazardous-area
- applications MIL-STD environmental testing
- requirements Customer-specific testing and qualification standards

Certifications and third-party testing can be undertaken based on product application, destination country and customer requirements.
Partner with Mecwin

From Agriculture to Aviation, Mecwin delivers engineered motor solutions for global applications. Design in India. Manufacture in Bengaluru. Supply to the World.

CONTACT US

Our Service & Support team are directly accessible in 18+ states in India.
MECWIN's trained service engineers can help with site survey, system sizing, installation, commissioning and on time service support.





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